

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Strong CS fundamentals: **kernel-level systems programming in C** (virtual memory, page replacement, kernel syscalls in Xinu OS), **concurrency and locking** (PostgreSQL row-level locking under concurrent load, Testcontainers-verified), and **performance engineering** (25% page fault reduction, 2,000 async req/sec). Production backend experience across distributed systems and data pipelines. Algorithms: NDCG@10, MRR@10, bootstrap significance testing.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Owned full backend of a production platform -- shipped 10+ features using **TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Redis**, Temporal.io workflow orchestration, Auth0, Docker, and Sentry; designed real-time event pipelines and async worker architecture serving real users.
- Built multi-model AI agent pipelines across 3 providers and 8+ models (Claude, GPT-4o, Gemini) via MCP; document processing pipelines (PDF gen, email parsing, S3); reviewed PRs across the engineering team.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built AI-driven claim automation (OCR, NLP, ML) improving accuracy 25%; automated AWS and GCP deployments with Grafana + Sentry observability.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Kafka + Spark + Hadoop distributed pipelines; cut processing time 15% and latency 20%.

PROJECTS

Demand Paging -- Xinu OS 2024

Kernel-level virtual memory in C for Xinu OS: 8-store paging, Second-Chance page replacement, inverted page tables, custom fault handler. Added 7+ kernel syscalls; reduced page faults **25%**, memory performance up **15%**.

C, Xinu OS, Virtual Memory, Kernel Systems, Page Replacement Algorithms

Inventory Reservation Service 2026

Transactional inventory API preventing overselling via **PostgreSQL row-level locking (SELECT FOR UPDATE)** across multi-SKU reservations. **Concurrency verified**: two simultaneous reservations against qty 1 produce exactly one success + one 409 Conflict. Idempotent endpoints, Flyway migrations, Docker, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Testcontainers, Docker, GitHub Actions

VectorLift -- Semantic Search & Ranking Engine 2026

Production semantic search on MS MARCO: BM25 baseline (Elasticsearch), dense bi-encoder retrieval (sentence-transformers, in-batch negatives), cross-encoder re-ranker. Evaluated **6 pipeline configs** with NDCG@10, MRR@10, MAP, and paired bootstrap significance testing; per-stage latency tracking via FastAPI.

Python, PyTorch, Sentence-Transformers, Elasticsearch, Qdrant, Kafka, FastAPI, Docker

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

Operating Systems, Algorithms, AI, Data Science, Software Engineering, Object Oriented Development

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | DS & Algorithms, OS, Big Data Infra, Machine Learning, Data Warehousing

SKILLS

Languages: C, Python, TypeScript, Java, JavaScript, SQL

Systems: Virtual Memory, Page Replacement, Kernel Syscalls, Concurrency, Row-Level Locking, Distributed Systems

Stack: Node.js, Spring Boot, FastAPI, PostgreSQL, Redis, Kafka, Docker, Temporal.io, GitHub Actions

Infra: AWS, GCP, Kubernetes, Prometheus, OpenTelemetry, Sentry, Elasticsearch, Qdrant

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)